

Graphs as Adjacency Processes

A Point Process Framework for Local Limits of Random Graphs

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Based on bachelor thesis work at TU/e;
Upcoming joint work with Pim van der Hoorn

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Motivation

Random graph limits are **hard**.

Point process limits are **easy** (relatively).

1. Associate every spatial random graph G with a point process ξ ;
2. Under reasonable conditions, lift point process convergence

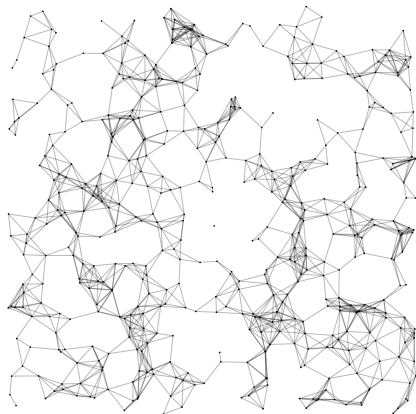
$$\xi_n \Rightarrow \xi_\infty$$

to a meaningful type of local graph convergence

$$G(\xi_n) \Rightarrow G(\xi_\infty).$$

Background - Spatial Random Graphs in $\mathbb{R}^d$

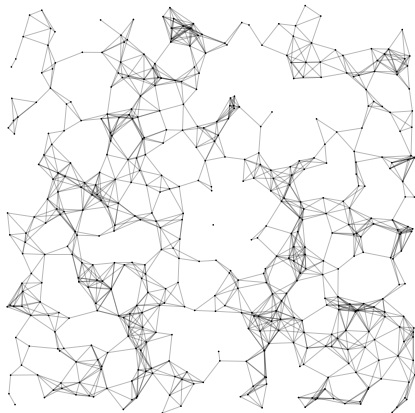
Random Geometric Graph



$$\mathbb{P}(E_{x,y}) = \mathbf{1}\{\|x - y\| \leq \rho\}$$

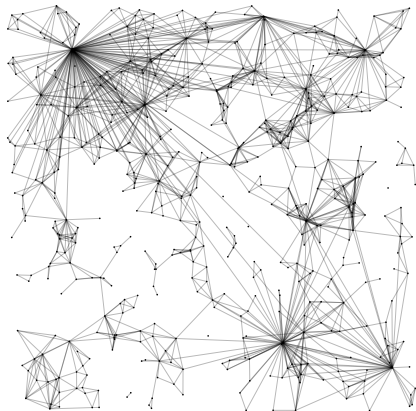
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SIRG/GIRG



$$\mathbb{P}(E_{x,y}) = \kappa(\|x - y\|, w_x, w_y)$$

Local Weak Convergence

Recall from yesterday, "local" convergence of rooted graphs $(G_n, p_n) \rightarrow (G_\infty, p_\infty)$. For all $r \in \mathbb{N}$, there exists some $N \in \mathbb{N}$, such that

$$B_r(G_n, p_n) \cong B_r(G_\infty, p_\infty), \quad \forall n \geq N.$$

This convergence defines the Benjamini-Schramm topology $\mathcal{G}^*$ on locally finite undirected rooted simple graphs. *Weak convergence* in this topology is then

$$(G_n, p_n) \Rightarrow (G_\infty, p_\infty).$$

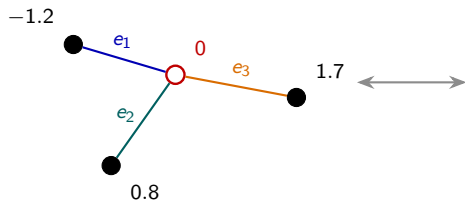
We call it *local weak convergence* when G_n are a.s. finite and the root is uniform, denoted

$$(G_n, U_n) \Rightarrow (G_\infty, p_\infty)$$

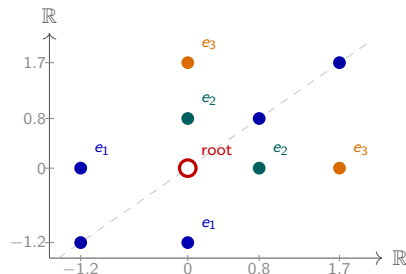
The Setup

- Represent a random graph G on vertices in $\mathbb{R}^d$ as an *adjacency process* in $\mathbb{R}^{2d}$

Rooted Labelled Graph $G(\xi, p)$



Symmetric Adjacency Measure (ξ, p)



- Can convergence of these point processes $\xi_n \Rightarrow \xi_\infty$ be lifted to local weak convergence of the corresponding graphs $G(\xi_n, U_n) \Rightarrow (G_\infty, p_\infty)$?

Constructing Adjacency Processes

Recall SIRGs:

- ▶ Constructed from a *collection of vertices* a proper metric space $\mathbb{X}$;
- ▶ Each pair connected with an isometric connection rule $\kappa(d_{\mathbb{X}}(x, y), w_x, w_y)$.

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To make our adjacency process ξ represent a SIRG:

- ▶ Formalize this collection of vertices as a point process Γ on $\mathbb{X}$;
- ▶ Turn κ into a measurable kernel C mapping to point processes on $\mathbb{X}^2$;
- ▶ Then $\xi \sim C(\Gamma)$ is the adjacency process.

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If this C is nice* and continuous-ish*, we can leverage

$$\Gamma_n \Rightarrow \Gamma_\infty \quad \implies \quad \xi_n \Rightarrow \xi_\infty$$

where $\xi_n \sim C(\Gamma_n), \xi_\infty \sim C(\Gamma_\infty)$

The Classical Case - Binomial in $\mathbb{R}^d$

If

- ▶ $\Gamma_n = \sum_{i=1}^n \delta_{X_i}$ with $X_i \stackrel{\text{i.i.d.}}{\sim} \text{Unif}(\mathbb{T}_n^d)$,
- ▶ C uses a SIRG rule $\kappa(\|x - y\|, w_x, w_y)$,
- ▶ $\kappa(t, \cdot, \cdot) \propto t^{-\alpha}$, where $\alpha > d$,

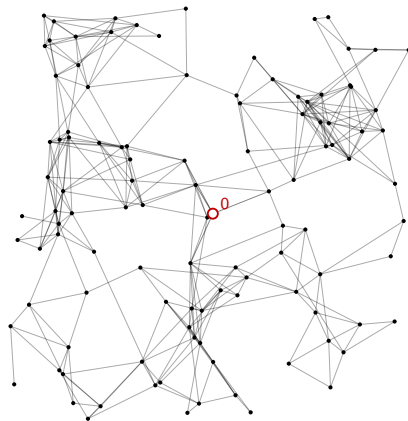
then

$$G(\xi_n, U_n) \Rightarrow G(\xi_\infty, 0),$$

where $\xi_n \sim C(\Gamma_n)$, $\xi_\infty \sim C(\Gamma_\infty)$.

Note also that $\Gamma_n^0 \Rightarrow \Gamma_\infty \dots$

Seems like Palm distribution $\sim$ uniform root choice.



Poisson SIRG limit

$$\Gamma_\infty \stackrel{d}{=} \text{PPP}(1) + \delta_0$$

The Main Theorem

Consider a sequence of finite stationary *vertex* processes $(\Gamma_n)_{n \in \mathbb{N}}$, placed along a nice* compact exhaustion of a proper metric space $\mathbb{X}$, and let C be a nice*, continuous-ish, isometric connection rule (measurable SIRG kernel). Suppose that

1. $\Gamma_n^o \Rightarrow \Gamma_\infty$;

2. The vertices concentrate via

$$\frac{|\Gamma_n|}{\mathbb{E}|\Gamma_n|} \xrightarrow{L^1} 1;$$

3. $(\xi_n, o)_{n \in \mathbb{N}} \sim C(\Gamma_n^o, \cdot)$ is *radially tight**.

Then, the corresponding graphs converge locally weakly,

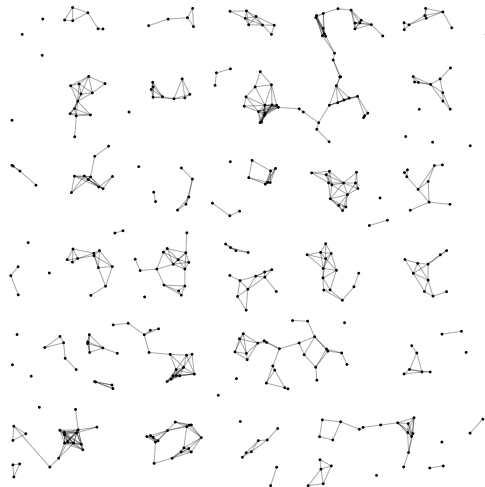
$$G(\xi_n, U_n) \Rightarrow G(\xi_\infty, o),$$

where $\xi_\infty \sim C(\Gamma_\infty, \cdot)$.

Novel Results - More Vertex Processes

Sticking with $\mathbb{R}^d$ and $\mathbb{T}_n^d$ for now, we can handle:

- ▶ Mixed-Binomial vertex processes
 $\sum_{n=0}^N \delta_{U_i}$
- ▶ Cox vertex processes (Phase-Shifted, Shot-Noise, Lévy)
- ▶ Neyman-Scott / Clustering
- ▶ Bernoulli Site processes (for atomic spaces)

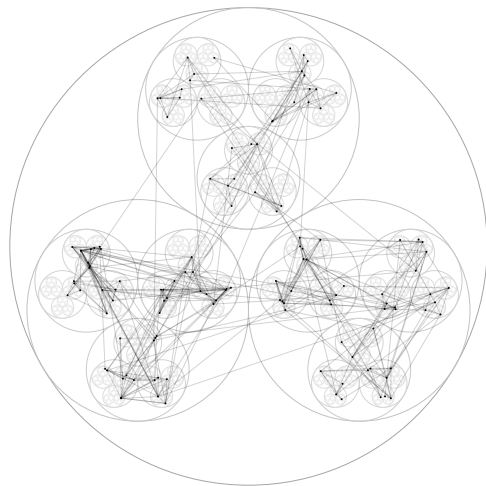


Soft RGG with Cox vertices

Novel Results - Non-Euclidean Spaces

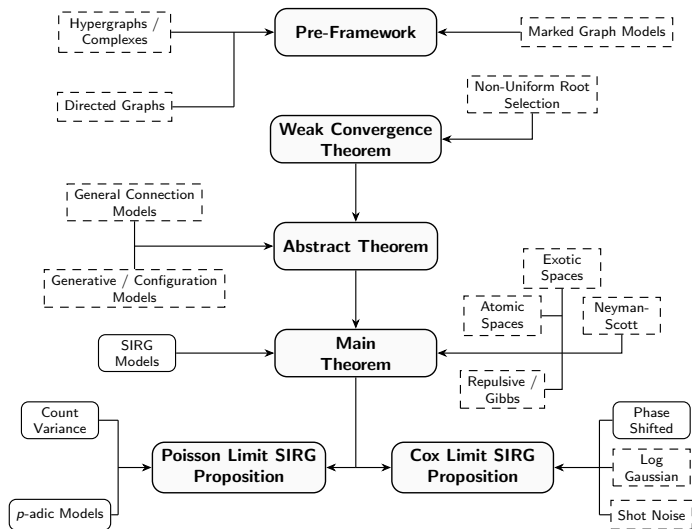
We can also handle the following approximating regimes:

- ▶ $B_{\mathbb{Q}}(0, p^n) \rightarrow \mathbb{Q}_p^d$
- ▶ Proper metric spaces that admit homogeneous quotient spaces, incl. infinite trees, graphs, and non-archimedean fields
- ▶ $\mathbb{H}^d$? Not yet... no homogeneous quotient spaces



SIRG in $\mathbb{Q}_3$

How YOU Can Benefit 🍷

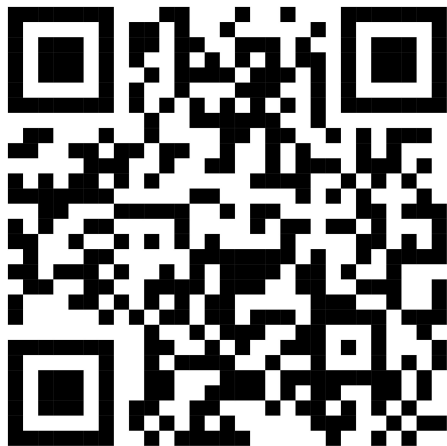


Overview of the main convergence results and model families feeding into them. Dashed borders indicate topics not directly explored in this paper.

Next Steps

- ▶ Preprint/Paper in the works with Pim van der Hoorn
- ▶ Directed graphs may be included in this revised framework
- ▶ Formalize conditions for more vertex processes (Cox, Neyman-Scott, ...)
- ▶ Hyperbolic SIRGs, eventually...

Scan Me (Thesis, Related Notes, Notes on Presentation Slides)



The Pre-Main Theorem

Consider a sequence of finite stationary adjacency processes $(\xi_n)_{n \in \mathbb{N}}$, placed along a nice* compact exhaustion of a proper metric space $\mathbb{X}^2$. Suppose that

1. $\xi_n^{(o,o)} \Rightarrow \xi_\infty$ (ξ_∞ will then also have a point at (o, o) almost surely);
2. The vertices concentrate via

$$\frac{|V(\xi_n)|}{\mathbb{E}|V(\xi_n)|} \xrightarrow{L^1} 1;$$

3. $(\xi_n^{(o,o)}, o)_{n \in \mathbb{N}}$ is *radially tight** (vertices reachable from o contained in a compact subset of $\mathbb{X}$ with high probability).

Then, the corresponding graphs converge locally weakly,

$$G(\xi_n, U_n) \Rightarrow G(\xi_\infty, o),$$

and ξ_∞ acts as a nice, sampleable representation of the local weak limit.